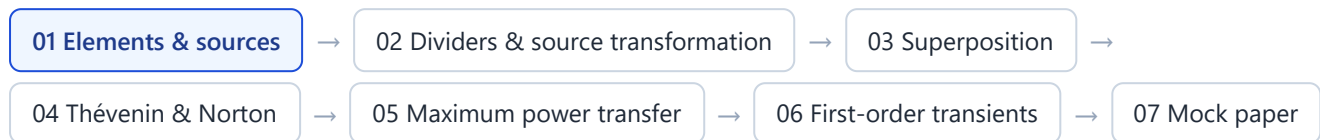


Circuits & Systems — Study Pack

ECE2107 · B.Tech ECE Sem 3 · built from the professor's class (notes to 12 Sept 2026) · every answer checked by circuit simulation

The path through this pack



Each file needs the ones before it. 04 uses 03's "switch a source off" rule and 01's dependent sources. 05 is built on 04. 06 uses 04 for the time constant.

File	What it teaches	Professor's questions in it
01	active vs passive, independent and dependent sources, the R, L, C laws, what can't change suddenly	none yet (theory)
02	voltage division, current division, source transformation, series–parallel reduction	4
03	superposition: method, and the cases where it fails	2
04	Thévenin and Norton, with a dependent source	1
05	maximum power transfer, efficiency, conjugate matching	2 sets
06	$0^- / 0^+ / \infty$ analysis, time constant, the one first-order formula	3
07	all of the above as one paper	13

Where the questions come from

Every question that opens a section is **quoted from the professor's board**, as written in the class notes, and tagged with the notes page: [Class p.7](#). No question is invented.

	Base	Status
P1	The professor's board questions (class notes, 23 pages)	13 questions , all used
P2	Textbook end-of-chapter problems	empty for now : no course hand-out yet, so the prescribed book is unconfirmed. The notes name Alexander & Sadiku, Van Valkenburg, A.K. Chakraborty

Marks per question: **not known** (no hand-out, no past paper yet).

How each file runs

Part	What happens
Map	the file's pieces, in order
Build	each step opens with a real question → you guess → the answer teaches the idea → a Check
Exam form	what to reproduce in the exam: statements, procedures, the formula sheet
Attempt	the professor's questions again, now answerable
Traps	the mistakes that cost marks
Self-test, Answers	answers are on the last page(s) of every file

THE ONE RULE THAT MAKES THIS WORK

When you reach a **Guess first** box, write a guess on paper **before** reading on: one line, however wrong. A wrong guess that gets corrected sticks better than a right answer you only read.

Mark every Check and Self-test item:

- **clean** → revisit in 3 days
- **with struggle** → revisit tomorrow
- **wrong** → redo today, after the file

Units and symbols used everywhere

- Uppercase **V, I** = DC (constant) values. Lowercase **v, i** = values that change with time.
- Resistances in **kΩ** with currents in **mA** give voltages in **V** directly: $2 \text{ k}\Omega \times 3 \text{ mA} = 6 \text{ V}$. Most questions use this.
- **R_{Th}**, **V_{Th}** = Thévenin resistance and voltage. **I_N** = Norton current. **R_L** = load resistance.
- **0⁻** = just before the switch moves. **0⁺** = just after. **∞** = long after (steady state).

How this pack was checked

- Facts: checked against the textbook-sourced circuits knowledge base (Hayt, Van Valkenburg, Sudhakar & Shyammohan) and Alexander & Sadiku.
- Numbers: every circuit in this pack was solved a second time by computer (nodal analysis and time simulation), and the answers here match.
- Where the class notes had a slip of the pen, the pack uses the corrected version and says so in a Note.